Adaptive Hybrid Control Error for Vehicle Platoons via Reinforcement Learning

Mid-Report

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1 Introduction

1.1 A Short Description

The modernization of intelligent transportation systems (ITS) has necessitated a paradigm shift from individual vehicle autonomy to cooperative automation. Vehicular platooning offers a promising solution to traffic congestion, fuel inefficiency, and road safety.

Thus, our work motivates an "Adaptive Hybrid Control strategy" using RL to adjust the weight between micro and macro. This transition from the standard Adaptive Cruise Control (ACC) to Cooperative Adaptive Cruise Control (CACC) introduces challenges in string stability, the ability to attenuate disturbances, especially under communication delays.

1.2 Motivation

A fundamental trade-off exists in platoon control architectures:

- **Local Control:** Relies on local sensors (radar/LiDAR). It is robust to network failures and ensures collision avoidance but is prone to string instability in long platoons.
- **Global Control:** Relies on V2V data from the leader. It provides excellent string stability but is highly sensitive to communication delays and packet loss.

1.3 Our contributions

- Modeling mixed traffic with human-driven(HDVs) and connected autonomous vehicles (CAVs), improving capacity and stability, in addition to extending traffic analysis from open platoons to closed ring-road systems.
- Proposing a mesoscopic RL framework combining micro- and macro-level traffic states with incorporated resistance forces on our input to create the nearest real life simulation.

2 Traffic Environment

2.1 Driver Models

2.1.1 Intelligent Driver Model (IDM)

The Intelligent Driver Model (IDM) is a deterministic microscopic car-following model describing the longitudinal dynamics of vehicle i as a function of its own velocity, the spacing to its predecessor, and their relative velocity [1]. The acceleration of vehicle i is defined as

$$\dot{v}_i = a_{max}\left[\underbrace{1-\left(\frac{v_i}{v_d}\right)^{\delta}}_{\text{free-road acceleration}} - \underbrace{\left(\frac{s^*(v_i,\Delta v_i)}{s_i}\right)^2}_{\text{interaction (braking) term}}\right]. \tag{1}$$

v_i: velocity of vehicle i,
v_d: the desired free-flow velocity
a_{max}: the maximum acceleration
δ: an acceleration exponent controlling how smoothly the vehicle approaches the desired velocity in free traffic.

When $v_i \ll v_0$, the free-road acceleration $\simeq 1$, so the vehicle accelerates approximately with magnitude a_{max}. As v_i approaches v_0, the term smoothly converges to zero, ensuring bounded acceleration and asymptotic convergence to the desired free-flow equilibrium [1].

The interaction term dominates when the actual spacing s_i is less than the desired spacing s^*, thereby inducing decceleration. The quadratic structure ensures strong braking for small gaps and guarantees collision-free behavior under realistic parameter choices [1].

The spacing is defined as

$$s_i = x_{i-1} - x_i - L, \tag{2}$$

where x_i and x_{i-1} denote the longitudinal positions of vehicle i and its leader, respectively, and L is the vehicle length. The relative velocity is defined as $\Delta v_i = v_i - v_{i-1}$.

The desired dynamic spacing is given by

$$s^*(v_i, \Delta v_i) = \underbrace{s_0}_{\text{jam distance}} + \underbrace{v_i T_H}_{\text{time-headway term}} + \underbrace{\frac{v_i \Delta v_i}{2\sqrt{a_{max} b}}}_{\text{anticipation term}} . \tag{3}$$

The constant term s_0 represents the minimum spacing in standing traffic. The linear term $v_i T_H$ reflects the constant time-headway policy characteristic of human driving behavior. The third term, proportional to $v_i \Delta v_i$, introduces anticipative braking by accounting for relative velocity differences and the maximum acceleration and desired deceleration denoted respectively by a_{max} and b [1].

2.1.2 Stochastic Intelligent Driver Model (SIDM)

To account for perception errors, behavioral variability, and unmodeled disturbances, the IDM can be extended by introducing a stochastic acceleration component [2]. The stochastic IDM (SIDM) is formulated as

$$\dot{v}_i = \underbrace{f(s_i, v_i, v_{i-1})}_{\text{deterministic IDM}} + \underbrace{\xi_i(t)}_{\text{stochastic acceleration noise}} , \tag{4}$$

where $f(\cdot)$ denotes the deterministic IDM acceleration function and $\xi_i(t)$ is a stochastic process representing white acceleration noise.

The stochastic term satisfies

$$\langle \xi_i(t) \rangle = 0, \tag{5}$$

$$\langle \xi_i(t) \xi_j(t') \rangle = Q\, \delta_{ij}\, \delta(t - t'), \tag{6}$$

where Q denotes the noise intensity, δ_{ij} is the Kronecker delta ensuring independence between different vehicles, and $\delta(t-t')$ is the Dirac delta distribution representing temporal white noise [2].

Close to the string stability threshold, noise can generate spatiotemporal correlations and traffic oscillations resembling stop-and-go waves. Consequently, the SIDM provides a unified framework for analyzing deterministic instabilities and noise-induced traffic dynamics [2].

2.1.3 Connected Autonomous Vehicle (CAV) Model

For connected autonomous vehicles, we adopt the point-mass longitudinal dynamics described in [3]. The motion of vehicle i is governed by

$$\dot{x}_i = v_i, \qquad \dot{v}_i = u_i, \tag{7}$$

In a car-following configuration, the spacing and relative velocity are defined as

$$s_i = x_{i-1} - x_i - L, \qquad \Delta v_i = v_i - v_{i-1}. \tag{8}$$

CAVs implement an Adaptive Cruise Control (ACC) strategy based on a Constant Time Headway (CTH) policy. The desired spacing is given by

$$s_i^{\text{ref}} = s_0 + T_C v_i, \tag{9}$$

where s_0 denotes the standstill spacing and $T_C > 0$ is the selected time headway.

Defining the spacing error

$$e_{p,i} = s_i - s_i^{\text{ref}}, \tag{10}$$

its dynamics follow from

$$\dot{s}_i = -\Delta v_i, \qquad \dot{s}_i^{\text{ref}} = T_C \dot{v}_i = T_C u_i, \tag{11}$$

which yields

$$\dot{e}_{p,i} = -\Delta v_i - T_C u_i. \tag{12}$$

A linear ACC controller that regulates both spacing and velocity errors can therefore be written as

$$u_i = k_s(s_i - s_i^{\text{ref}}) - k_v \Delta v_i, \tag{13}$$

where $k_s > 0$ and $k_v > 0$ denote the spacing and velocity feedback gains, respectively. This control law constitutes a linear realization of the CTH policy and is consistent with the mesoscopic ACC formulation proposed in [3].

An important property of the CTH-based ACC strategy is its capability to guarantee weak string stability using only local measurements [4]. In the frequency domain, the disturbance propagation transfer function satisfies

$$\|H(j\omega)\|_\infty \leq 1, \tag{14}$$

ensuring that spacing disturbances are not amplified upstream along a vehicle string. Moreover, robustness with respect to sensing and actuation delays imposes a constraint on the admissible implementation lag. In particular, the total delay must satisfy

$$T_C \leq \frac{h_w}{2}, \tag{15}$$

indicating that allowable delay is directly limited by the chosen time headway [4].

2.2 Mixed Traffic Model

We consider a mixed traffic system composed of N vehicles circulating on a single-lane ring road of length L_{ring}. The vehicle indices satisfy periodic boundary conditions such that vehicle 1 follows vehicle N.

Let $x_i(t)$ and $v_i(t)$ denote the position and velocity of vehicle $i \in \{1, \ldots, N\}$. The ring-road constraint is given by

$$x_{i-1}(t) > x_i(t), \qquad x_0(t) := x_N(t) + L_{\text{ring}}. \tag{16}$$

The spacing is defined as

$$s_i = x_{i-1} - x_i - L, \tag{17}$$

where periodicity applies for $i = 1$.

Let $\mathcal{H}$ denote the set of HDVs and $\mathcal{C}$ the set of CAVs, with penetration rate

$$P = \frac{|\mathcal{C}|}{N}. \tag{18}$$

Human-driven vehicles are modeled according to the stochastic Intelligent Driver Model introduced in Subsection 2.1.2.

HDVs do not utilize acceleration information of the preceding vehicle, reflecting the absence of communication capabilities. Furthermore, HDVs are characterized by a larger effective reaction time T_H, representing the delay associated with human perception and actuation. Consequently, their response to traffic perturbations is slower and more susceptible to amplification of stochastic fluctuations.

Connected autonomous vehicles are modeled according to the ACC framework introduced previously in 2.1.3. In contrast to HDVs, CAVs are assumed to exhibit higher sensitivity coefficients and a shorter desired time headway T_C, reflecting their faster sensing [5]. The condition $T_C < T_H$ therefore captures the higher responsiveness of CAVs compared to HDVs.

When a CAV follows an HDV, communication-based acceleration information is unavailable. In this degraded configuration, the feedforward gain is set to $k_a = 0$, and the effective time headway is increased to $T_C^{\text{deg}} > T_C$, representing reduced cooperative capability. This modeling choice captures the performance degradation of CAVs in mixed traffic reported in [5] and [6].

The closed-loop mixed system is therefore

$$\dot{v}_i = \begin{cases} f_{\text{IDM}}(s_i, v_i, v_{i-1}) + \xi_i(t), & i \in \mathcal{H}, \\ k_s(s_i - s_i^{\text{ref}}) - k_v \Delta v_i + k_a a_{i-1}, & i \in \mathcal{C}. \end{cases} \tag{19}$$

2.3 Macroscopic Information

Up to this point, only microscopic vehicle interactions have been considered. To improve the performance of the proposed framework, macroscopic traffic information are introduced according to [3]. In particular, we define the mean velocity on the ring as

$$\bar{v}(t) = \frac{1}{N} \sum_{i=1}^{N} v_i(t), \tag{20}$$

and the velocity variance

$$\sigma_v^2(t) = \frac{1}{N} \sum_{i=1}^{N} \left(v_i(t) - \bar{v}(t)\right)^2. \tag{21}$$

The mean velocity represents the macroscopic traffic state, while the variance quantifies the level of speed dispersion and therefore the intensity of oscillations. As discussed in [3], the variance provides a compact indicator of transient traffic behavior and string instability.

3 Reinforcement Learning

In this project, reinforcement learning (RL) is used to improve the behavior of connected and automated vehicles (CAVs) in mixed traffic. The main idea is not to replace the baseline

controller, but to add an adaptive layer on top of it. This learned layer adjusts the time headway policy according to the current traffic situation. In this way, the controller keeps its physical interpretation while gaining the ability to react to complex traffic patterns that are difficult to model explicitly.

3.1 Fundamentals

The RL problem is formulated as a Markov Decision Process (MDP),

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r),$$

where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, P represents the traffic dynamics, and r is the reward.

At each time step, the agent observes the traffic state, selects an action, applies it to the controller, and receives a reward. In the current implementation, the state of each CAV i is defined as

$$x_i = (\mu_{v,i}, \sigma^2_{v,i}, \delta^{\text{macro}}_{v,i}, v_i, s_i, \Delta v_i, a_{\ell(i)}, \alpha_i^{prev}),$$

where $\mu_{v,i}$ and $\sigma^2_{v,i}$ are the upstream mean velocity and velocity variance, $\delta^{\text{macro}}_{v,i} = v_i - \mu_{v,i}$ is the speed mismatch with the local upstream flow, v_i is the ego speed, s_i is the spacing to the leader, Δv_i is the relative speed, $a_{\ell(i)}$ is the leader acceleration, and α_i^{prev} is the previously applied headway scaling factor.

This state combines *microscopic* information, such as spacing and relative velocity, with *mesoscopic* information, such as the average speed and speed variance of upstream traffic. This is important because local measurements alone may not be enough to detect the development of traffic waves.

The action is a residual correction of the headway scaling:

$$\mathcal{A} = \{\Delta\alpha_i \in [-\Delta\alpha_{\max}, +\Delta\alpha_{\max}]\}.$$

The final headway scaling factor is then given by

$$\alpha_i(t) = \text{clip}\left(\alpha_i^{rule}(t) + \Delta\alpha_i(t)\right),$$

where $\alpha_i^{rule}(t)$ is the baseline headway scaling provided by the rule-based mesoscopic module. Therefore, RL only refines the baseline decision instead of replacing it. The learned policy can be written as

$$\pi : \mathcal{S} \to \mathcal{A}, \qquad \Delta\alpha_i(t) = \pi(x_i(t)).$$

3.2 Reward design

The reward function is designed to encourage safe, smooth, and efficient traffic behavior while improving string stability. The total reward is

$$r_t = r_{\text{sf}}[t] + r_{\text{sp}}[t] + r_{\text{ef}}[t] + r_{\text{cf}}[t] + r_{\text{ss}}[t] + r_{\sigma^2}[t].$$

The desired spacing is defined by

$$s_i^{des}[t] = d_0 + h_i'[t]\, v_i[t],$$

and the spacing error is

$$e_i[t] = s_i[t] - s_i^{des}[t].$$

The different reward terms have a clear physical meaning:

- r_{sf}: safety term, penalizing unsafe spacing and too small time headway;
- r_{sp}: spacing regulation term, encouraging tracking of the desired spacing;
- r_{ef}: efficiency term, encouraging the vehicle speed to remain close to the upstream mean speed;
- r_{cf}: comfort term, penalizing high jerk;
- r_{ss}: string stability term, penalizing amplification of spacing errors along the platoon;
- r_{σ^2}: traffic smoothness term, penalizing large velocity variance in the traffic stream.

This reward formulation is useful because it combines several objectives into a single learning signal. It allows the RL agent to balance local vehicle behavior and global traffic quality.

3.3 PPO

Proximal Policy Optimization (PPO) is a policy-gradient RL algorithm that updates the policy in a stable and conservative way. Its main advantage is that it is simple to implement and usually robust during training. PPO avoids large policy updates by using a clipped objective:

$$L^{\mathrm{PPO}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right],$$

where

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{\mathrm{old}}}(a_t|s_t)},$$

and $\hat{A}_t$ is the estimated advantage.

For our problem, PPO is attractive because the action is continuous and the controller must remain stable during learning. It is a good candidate for a first implementation because of its simplicity and reliability.

3.4 SAC

Soft Actor-Critic (SAC) is another RL algorithm for continuous control. Unlike PPO, SAC is an off-policy method and includes an entropy term in its objective:

$$J(\pi) = \sum_t \mathbb{E}\left[r(s_t, a_t) + \beta \mathcal{H}(\pi(\cdot|s_t)) \right],$$

where $\mathcal{H}(\pi(\cdot|s_t))$ is the policy entropy and β controls the exploration level.

The main interest of SAC is its good sample efficiency and its strong performance in continuous-action problems. Since our action is a continuous residual correction $\Delta\alpha_i$, SAC is also well suited to this framework. However, compared with PPO, SAC is usually more complex to tune and explain. We also chose to consider SAC because the mixed-traffic environment is stochastic due to HDV behavior. SAC learns from a replay buffer containing past interactions, which allows it to reuse previous experience instead of relying only on newly collected data. This can make learning more sample-efficient and potentially more robust to traffic variability. In contrast, PPO uses mainly the data generated by the current policy at each training iteration. For this reason, PPO is often easier to implement, while SAC is an interesting alternative for noisy and stochastic environments.

3.5 Role of RL in the proposed framework

The role of RL in our approach is to make the controller adaptive while preserving the original control structure. The baseline controller still ensures a physically meaningful longitudinal behavior, while RL improves the headway adaptation using both microscopic and mesoscopic traffic information.

This integration has three main advantages. First, it keeps the controller interpretable for a control-theory audience. Second, it allows the CAV to react not only to its immediate leader, but also to collective upstream traffic conditions. Third, it avoids solving a complex optimization problem online at each time step. For these reasons, RL is a suitable tool for improving stability and smoothness in mixed traffic.

4 System Architecture

The proposed system is organized as a layered control architecture for mixed traffic on a ring road. Its main objective is to combine microscopic control with mesoscopic traffic awareness. The complete framework can be divided into four main parts: the simulation layer, the mesoscopic perception layer, the mesoscopic adaptation layer, and the control layer. Each part has a specific role, and together they allow the CAVs to react not only to their immediate leader but also to the collective traffic conditions upstream.

4.1 Overall structure

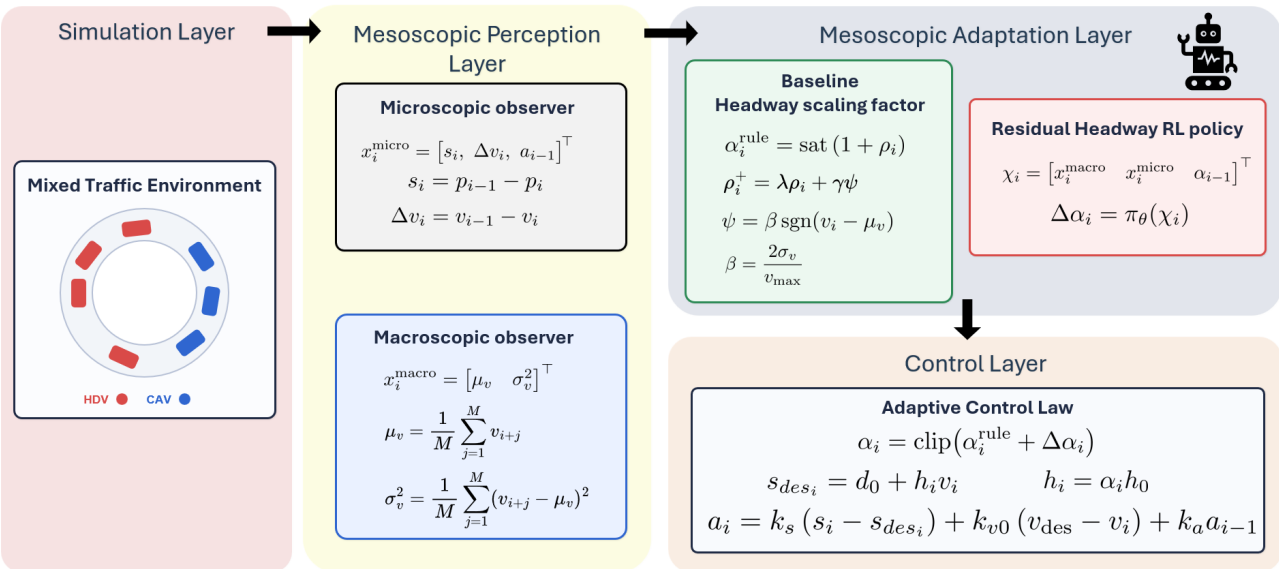

Figure 1: Overview of the proposed system architecture.

The simulation layer represents the mixed-traffic environment, where human-driven vehicles (HDVs) and connected and automated vehicles (CAVs) evolve on the same road. At each time step, the CAV receives observations from the traffic environment, processes them through two perception channels, computes an adaptive parameter, and finally applies the resulting control law. In this way, the architecture separates *observation*, *adaptation*, and *actuation*, which makes the framework easier to interpret from a control perspective.

4.2 Simulation layer

The simulation layer contains the traffic environment and generates the variables needed by the control framework. The road is modeled as a ring road, which avoids boundary effects and makes the appearance of traffic oscillations easier to study. HDVs and CAVs coexist in the same environment, but only the CAVs use the adaptive control architecture.

This layer provides the real-time traffic variables used by the upper layers, such as positions, velocities, spacing, and acceleration. It is therefore the source of both local and collective traffic information.

4.3 Mesoscopic perception layer

The perception layer extracts the information used by the controller. It contains two complementary observers.

Microscopic observer. The microscopic observer describes the local car-following situation of vehicle i. It uses the spacing, relative velocity, and ego speed:

$$x_i^{\text{micro}} = \begin{bmatrix} s_i & \Delta v_i & a_{i-1} \end{bmatrix}^\top,$$

with

$$s_i = p_{i-1} - p_i, \qquad \Delta v_i = v_{i-1} - v_i.$$

These variables describe the immediate interaction with the preceding vehicle and are the standard quantities used by longitudinal controllers.

Macroscopic observer. The second observer captures collective traffic behavior upstream of the CAV. Instead of looking only at the leader, it summarizes the traffic state over a perception horizon. The two main quantities are the upstream mean velocity and the upstream velocity variance:

$$x_i^{\text{macro}} = \begin{bmatrix} \mu_v & \sigma_v^2 \end{bmatrix}^\top,$$

with

$$\mu_v = \frac{1}{M}\sum_{j=1}^{M} v_{i+j}, \qquad \sigma_v^2 = \frac{1}{M}\sum_{j=1}^{M}(v_{i+j} - \mu_v)^2.$$

The mean velocity gives an estimate of the average upstream traffic speed, while the variance indicates how irregular or unstable the traffic stream is.

4.4 Mesoscopic adaptation layer

The adaptation layer transforms the observed traffic information into a headway modification for the CAV controller. It contains two parts: a baseline adaptive headway mechanism and a residual reinforcement learning policy.

Baseline adaptive headway. The baseline mechanism computes a rule-based headway scaling factor from the macroscopic traffic information. The update can be written as

$$\alpha_i^{\text{rule}} = \text{sat}(1 + \rho_i),$$

with the internal state

$$\rho_i^+ = \lambda \rho_i + \gamma \psi,$$

and the perception signal

$$\psi = \beta \,\mathrm{sgn}(v_i - \mu_v), \qquad \beta = \frac{2\sigma_v}{v_{\max}}.$$

This mechanism has a simple interpretation. When upstream traffic becomes more variable, the controller increases the effective headway in order to react more cautiously. When traffic is smoother, the headway can be reduced. The filtering through ρ_i also avoids abrupt parameter changes.

Residual RL policy. On top of the baseline adaptation, a residual policy computes a correction term:

$$\chi_i = \begin{bmatrix} \mu_v & \sigma_v^2 & (v_i - \mu_v) & v_i & s_i & \Delta v_i & a_{i-1} & \alpha_i^{prev} \end{bmatrix}^\top, \qquad \Delta\alpha_i = \pi_\theta(\chi_i).$$

The role of this block is to refine the rule-based decision without changing the overall structure of the controller. In other words, the adaptive law remains physically interpretable, while the learning block adds flexibility when the traffic situation becomes more complex. The resulting policy input therefore combines mesoscopic information, local car-following variables, and the previously applied headway state in a single 8-dimensional vector.

4.5 Control layer

The control layer converts the adapted headway into a longitudinal acceleration command for the CAV. The final scaling factor is

$$\alpha_i = \mathrm{clip}\big(\alpha_i^{\mathrm{rule}} + \Delta\alpha_i\big)\,,$$

and the corresponding time headway becomes

$$T_i = \alpha_i T_0.$$

This adapted headway is then used inside the microscopic CACC/CTH controller:

$$a_i = k_s\,(s_i - d_0 - T_i v_i) + k_v \Delta v_i + k_a a_{i-1} + k_{v0}(v_{\mathrm{des}} - v_i).$$

This equation shows clearly how the architecture is organized. The lower-level controller remains a classical feedback law based on spacing, relative speed, leader acceleration, and desired speed regulation. The upper layers do not replace this controller; they only adapt the time-headway parameter T_i used inside it.

5 Current Progress, Results and Perspectives

This section reports the state of the consolidated implementation after the first report draft. From this point onward, the project is presented as a single main codebase. The experiments below correspond to the current pipeline only: the same residual RL controller, the same mesoscopic adaptation mechanism, and the same evaluation scripts used to produce the latest local results.

5.1 What was added after the first draft

Several substantial changes were applied after the initial report version and before the final experiments presented here:

- the missing evaluation metrics were integrated into the main pipeline, in particular the late-run objective `mean_speed_last100`, string-stability validity flags, collision-clamp counts, and safety-constraint summaries;
- a checkpoint-aware early-stopping monitor was introduced, together with best-model restoration and explicit summaries of the best observed training point;
- the training loop was rewritten into a vectorized PyTorch backend able to batch several environments at once and run on GPU while keeping the same 8-feature observation format used by the main controller;
- the report-generation scripts were hardened so that evaluation outputs now include the effective configuration and the comparison report no longer breaks when a metric is unavailable or invalid.

These updates were not only implementation cleanups. After they were applied, additional training and evaluation runs were launched again, so the results reported below correspond to the updated pipeline rather than to the earlier draft state.

5.2 Experimental protocol used for the updated results

The updated experiments were run on the CentraleSupélec DCE infrastructure with a CUDA-enabled PyTorch environment and an NVIDIA RTX 2080 Ti GPU. The GPU acceleration was used for training only; the evaluation runs still use the standard simulation path so that the learned models are tested in the same operational setting as the controller used at inference time.

The final experimental campaign contains:

- two training runs: one PPO run and one SAC run, both at 50 % human-driver ratio;
- 200 physical training episodes per algorithm;
- 8 parallel environments during training, which means that each 200-episode run is executed as 25 vectorized training batches;
- nine evaluation runs in total: No RL, PPO, and SAC, each tested at 25 %, 50 %, and 75 % human-driver ratios.

The evaluation focuses on the metrics that are the most informative for this project: late-run mean speed, late-run speed variance, comfort, minimum gap, collision warnings, collision clamps, and the validity of the string-stability experiment.

5.3 Training summary

Table 1 summarizes the two updated training runs. The important point is that the best observed training point appears early for both algorithms, whereas the final checkpoint is not the best one in terms of variance minimization.

Table 1: Training summary for the updated GPU-enabled pipeline.

Method	Physical episodes	Batches	Best batch	Best variance	Final variance
PPO	200	25	9	1.471	1.877
SAC	200	25	3	1.530	1.762

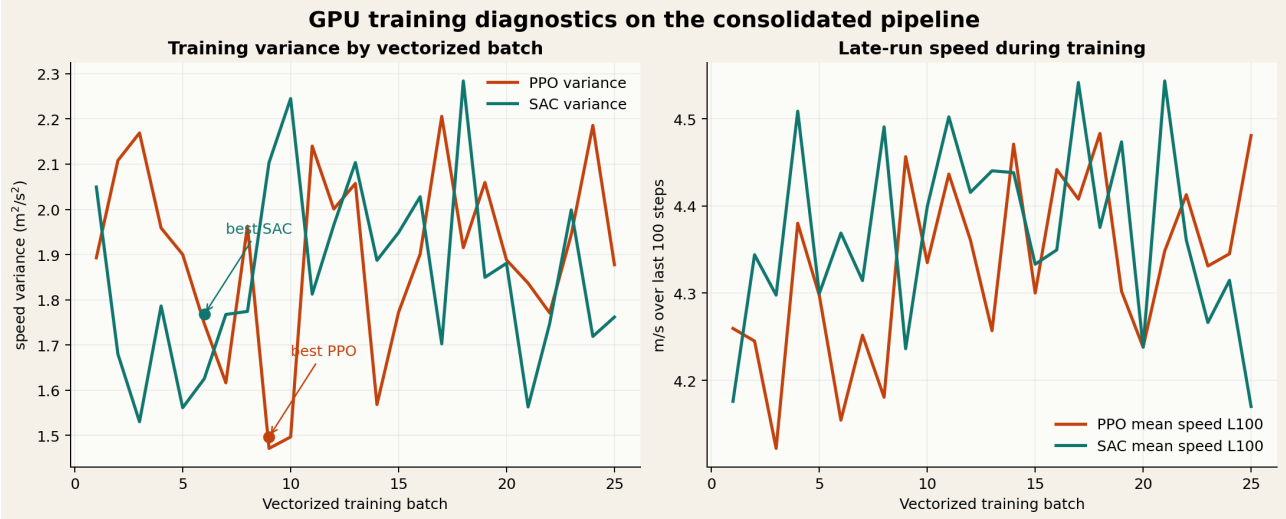

Figure 2: Training diagnostics from the final PPO and SAC runs. The horizontal axis counts vectorized batches, not single episodes; with 8 parallel environments, 25 batches correspond to 200 physical episodes.

Two remarks are worth emphasizing. First, PPO took noticeably longer than SAC in wall-clock time. This is expected in the current configuration because PPO performs many more optimizer updates per collected rollout batch, whereas SAC uses a smaller off-policy update loop. Second, the early-stopping monitor did not actually stop either run because the patience remained too large for a 25-batch training campaign. This is informative in itself: the monitoring logic is now present, but its configuration still needs to be tightened if it is to play a meaningful role.

5.4 Evaluation results

Table 2 reports the final comparison across the three human-driver ratios. The most informative quantities are the late-run objective `mean_speed_last100` and the late-run variance `speed_var_last100`, because they focus on the stabilized part of the trajectory rather than averaging the whole transient.

Table 2: Updated evaluation results for the consolidated pipeline.

HR	Method	Mean speed$_{100}$ (m/s)	Var$_{100}$ (m^2/s^2)	RMS acc. (m/s^2)	RMS jerk (m/s^3)	Min gap (m)	Warnings	Clamps
25 %	No RL	**5.175**	0.0098	**0.578**	**8.682**	**0.510**	**0**	3
	PPO	4.974	0.0185	0.679	10.334	0.510	**0**	3
	SAC	5.164	**0.0063**	0.635	9.632	0.510	**0**	3
50 %	No RL	4.716	1.142	**0.437**	**5.169**	**0.510**	**0**	3
	PPO	4.717	0.993	0.451	5.530	**0.510**	**0**	3
	SAC	**4.719**	**0.968**	0.454	5.577	**0.510**	**0**	3
75 %	No RL	**2.887**	3.718	**0.420**	**3.451**	0.307	42	4
	PPO	2.880	**3.666**	0.421	3.532	**0.345**	46	**3**
	SAC	2.866	3.777	0.424	3.531	0.300	**39**	5

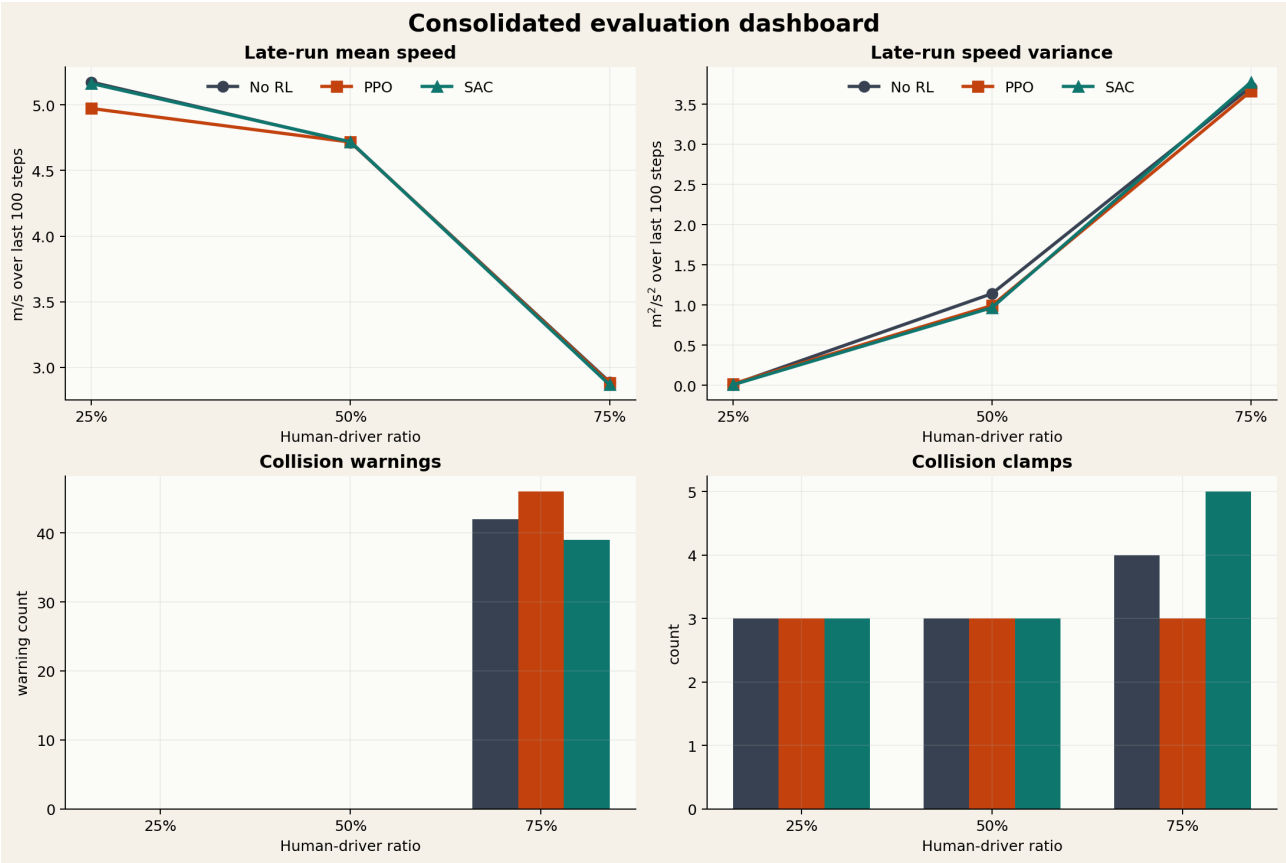

Figure 3: Self-contained evaluation dashboard stored locally in the report folder. The four panels show the main compromise between throughput, smoothness, collision warnings, and collision clamps.

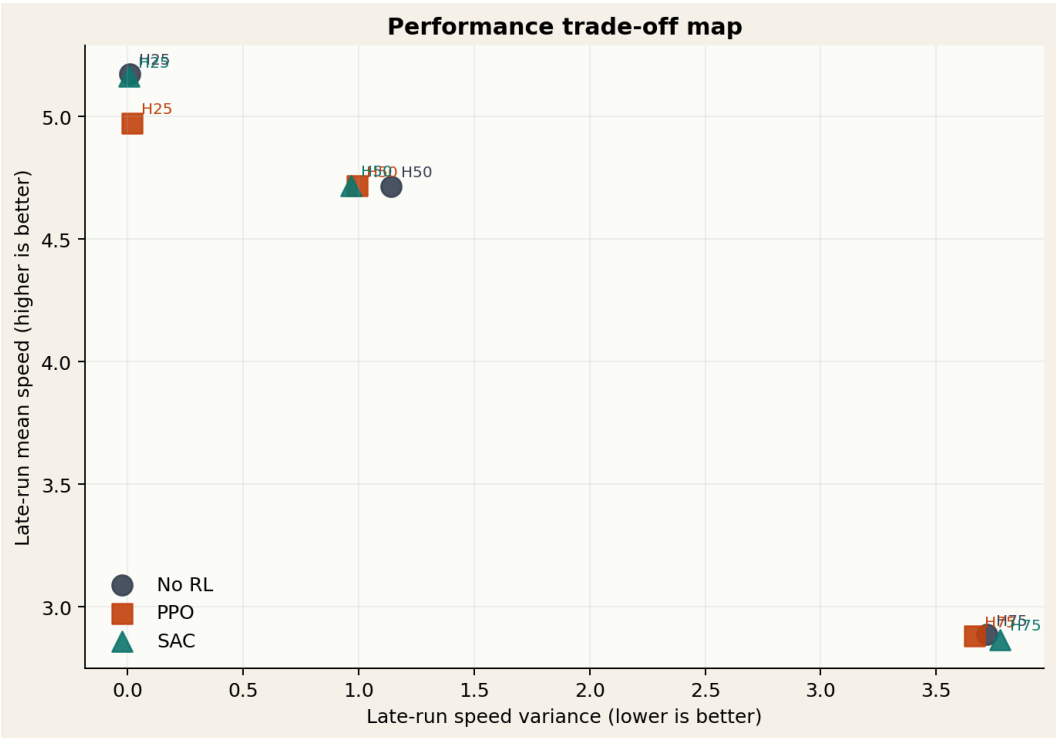

Figure 4: Trade-off map using late-run mean speed and late-run speed variance. The 50 % human-driver regime is the clearest region where learning improves the throughput–smoothness compromise.

5.5 Reasoned interpretation

Table 3 summarizes the practical reading of the results. It is intentionally qualitative: the purpose is not to crown a universal winner, but to identify where each method is actually useful and where the current limitations remain dominant.

Table 3: Qualitative synthesis of the current results.

Regime	Current best reading	Main reason
25 % humans	No RL remains the strongest overall baseline	The traffic is already smooth; learning does not bring a meaningful throughput gain, and PPO even degrades comfort. SAC lowers variance slightly, but without a safety gain.
50 % humans	SAC has the best current compromise, with PPO close behind	This is the only regime where RL clearly improves late-run variance while preserving mean speed. The SAC run reduces late variance by about 15 % relative to No RL.
75 % humans	No method is satisfactory yet	The regime remains dominated by safety events. PPO slightly improves variance and clamp count, SAC slightly reduces warning count, but every method still suffers collision clamps.

The main conclusion is that the 50 % human-driver regime is the real sweet spot of the current approach. At 25 %, the baseline controller is already so stable that learning has little room to help. At 75 %, the stochasticity introduced by the human-driven vehicles is strong enough that the learned residual policy still fails to make the system consistently safe. In between these two extremes, however, RL manages to reduce oscillations without sacrificing throughput, which is precisely the behavior the project aims to obtain.

Another important point is that all current runs fail the formal safety summary. This is not a bug in the evaluation code. The safety criterion checks both the minimum-gap threshold and the absence of collision clamps, and every run violated at least one of these conditions. The same observation explains why the string-stability amplification ratio is reported as invalid in the latest results: once collision clamps occur, the perturbation experiment is no longer a clean string-stability test.

A limitation that must be stated explicitly. The current results are based on a single random-seed evaluation campaign. They are therefore valid as matched-condition comparisons, but they are not yet sufficient to claim genuine generalization or to rule out over-training. The fact that both algorithms reached their best training variance early and then drifted away from it is a useful warning sign: longer training is not automatically better, and the next credible step is to evaluate the saved checkpoints on unseen seeds.

A remark I found particularly interesting. The most interesting pattern in these runs is that the learning signal seems to be genuinely informative only in the intermediate regime. This is a pleasant surprise from a traffic-flow point of view: the controller is not simply "better everywhere" or "worse everywhere". Instead, it helps most when the traffic is neither too easy nor too chaotic, which is exactly the kind of regime where hybrid control is supposed to matter.

5.6 Future perspectives

The most immediate next steps are now quite clear:

- evaluate the selected PPO and SAC checkpoints on several unseen random seeds to check whether the observed gains at 50 % human drivers persist outside the current run;
- reduce the early-stopping patience to a value that makes sense for a 25-batch vectorized training campaign;
- enforce safety more directly during learning, for example with stronger hard constraints or constrained-RL formulations, because soft penalties alone are not preventing collision clamps in the difficult regimes;
- tune PPO and SAC differently instead of using almost parallel training budgets, since their optimization dynamics are not comparable in wall-clock time;
- extend training beyond a single human-driver ratio, either through curriculum training or mixed-ratio batches, in order to make the policy less specialized to the 50 % case.

At the present stage, the project already demonstrates a meaningful result: GPU-accelerated training makes larger experiments practical, and the learned residual controller can improve the throughput–smoothness trade-off in the intermediate mixed-traffic regime. The remaining challenge is no longer whether the idea can work at all, but how to make it robust and safe enough to support stronger scientific claims.

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